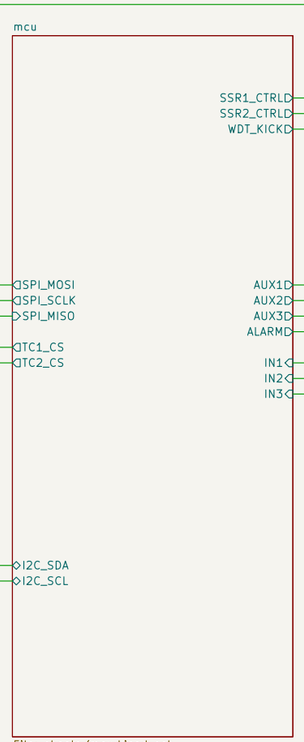
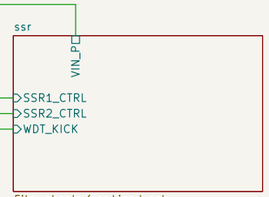
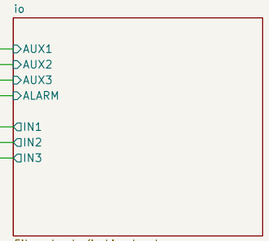








NOTES — Bisque kiln controller, rev B

SSR terminals J4/J9: pin1 = VIN_P (24 V, fused and TVS-clamped, ALWAYS LIVE), pin2 = the switched low side. The watchdog gates the shared return (Q7), not the supply, so a tripped watchdog leaves 24 V on pin 1 and opens pin 2. Boot-safe: R7/R20 hold each gate down while the optos are dark.
The optos are NOT an isolation barrier — both sides share GND. Rev B's opto-ISOLATED design was reverted and stays reverted; U8/U9 are here to keep a 24 V gate fault off the GPIO.
Per channel <=200 mA: F1 holds at 750 mA and U11 takes ~250.

TC1 terminal J3 / TC2 terminal J8: pin1 = K+, pin2 = K- — both float, biased near AGND only through each MAX31856's internal BIAS network. Ungrounded-junction probes required: two grounded-junction probes in one kiln would loop through it.

Nav switch J6 is panel-mounted; inputs use ESP32 pull-ups.

J11: IN1/IN2/IN3 (lid/gas-flow/spare) + GND — dry contact each channel to GND.

Display J5 (14-pin, ST7796S + XPT2046 touch module, LCDWIKI MSP4021): 1=+5V 2=GND 3=CS 4=RST 5=DC 6=SDI/MOSI 7=SCK 8=BL 9=SDO/MISO 10=T_CLK 11=T_CS 12=T_DIN 13=T_DO 14=T_IRQ. PIN 1 IS +5V, NOT 3V3 — the module regulates on board; do not wire a 3.3V-only panel to it. Logic is 3.3V; touch shares SPI2 with the LCD and both MAX31856s (R39-R43 damping).

J12: CT current inputs, CTA_P/CTA_N/CTB_P/CTB_N — one CT clamp per SSR zone.

J13: DNP SELV voltage sense header. NOT mains-rated, not fitted — do not wire to AC mains. Y1 is a packaged 3.579545 MHz oscillator (no load caps; C25 is its bypass) — see DESIGN-NOTES.md, Y1. ADE7953 I2C address 0x38 would collide with a PCF8574A added on J7/J14.

Design reasoning per block: hardware/kicad/DESIGN-NOTES.md.

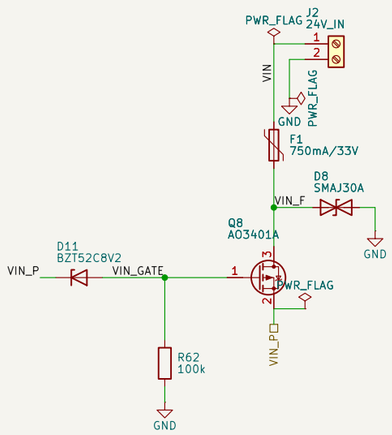
4-layer, 100 x 100 mm, JLCPCB standard process
ESP32-S3-WROOM-1U-N16R2 + 2x MAX31856 + dual SSR + ADE7953
Bisque project

Sheet: /
File: bisque-controller.kicad_sch

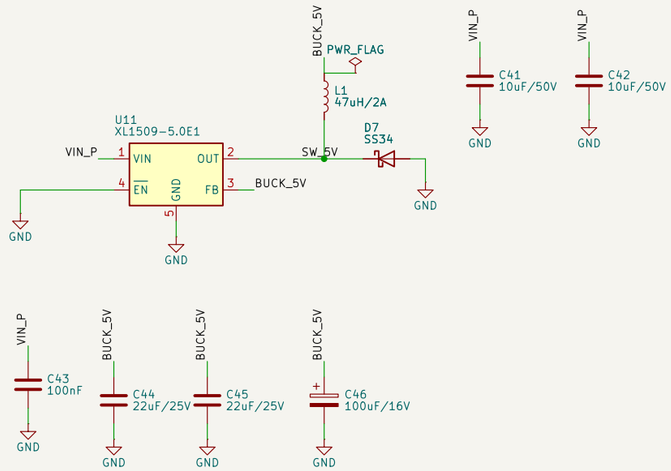
Title: Bisque Kiln Controller

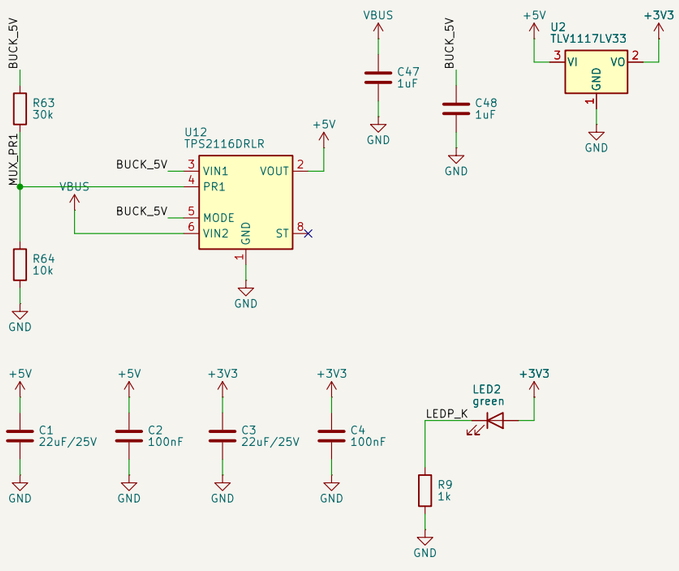
Size: A3	Date: 2026-10-22	**Rev: B.3**
KiCad E.D.A. 10.0.6		Id: 1/8

POWER IN 24 VDC terminal or USB
F1+D8 clamp-and-blow at the entry. Q8 blocks reverse polarity as a P-FET, not a diode: its body diode conducts on correct polarity and the gate then pulls itself on through R62, clamped by D11 because 24 V is twice the AO3401A's V_GS rating. U11 (right) makes BUCK_5V; U12 selects between it and USB VBUS to make +5V, giving the buck PRIORITY via R63/R64 rather than letting the higher voltage win -- a 5.25 V host would otherwise source the SSR terminals and the relay bank. U12 also blocks USB back-feeding the unpowered buck. C47/C48 are its input caps, one per channel, at the pins.
U2 drops +5V to +3V3



24 V -> 5 V BUCK XL1509-5.0, fixed output
C41-C43 in, D7 catch, L1 47uH, C44/C45/C46 out. FB ties to the output because the part is the fixed 5.0 V version; EN is active LOW and grounded. C46 is an ELECTROLYTIC on purpose: voltage-mode with fixed compensation needs its ESR zero, and an all-ceramic output has none. Keep C41/C42 - U11 - D7 tight: that loop, not L1, is what radiates.

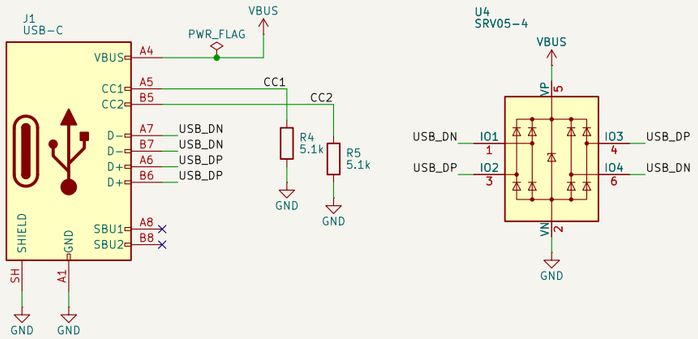




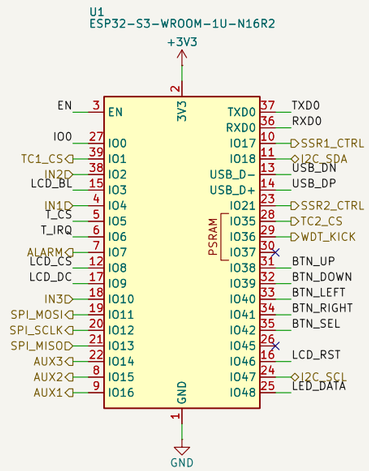
4-layer, 100 x 100 mm, JLCPCB standard process
ESP32-S3-WROOM-1U-N16R2 + 2x MAX31856 + dual SSR + ADE7953
Bisque project
Sheet: /power/
File: power.kicad_sch
Title: Bisque Kiln Controller - Power input and 24 V -> 5 V buck

Size: A3	Date: 2026-10-22	**Rev: B.3**
KiCad E.D.A. 10.0.6		Id: 2/8

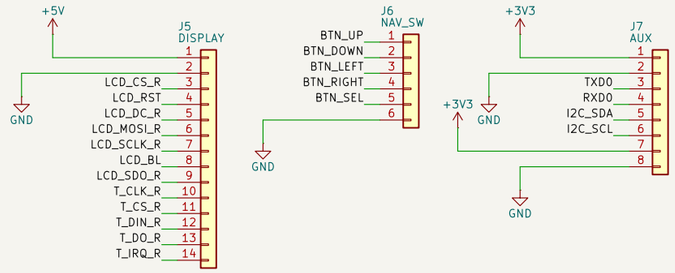
USB-C
native USB flashing + ESD



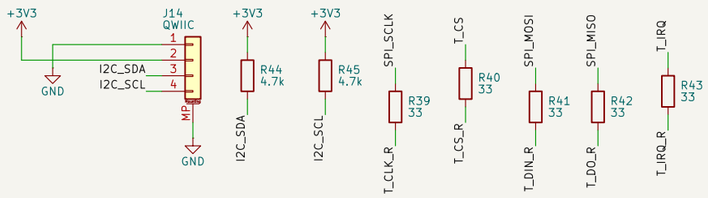
ESP32-S3-WROOM-1U-N16R2
GPIOs = firmware Kconfig defaults

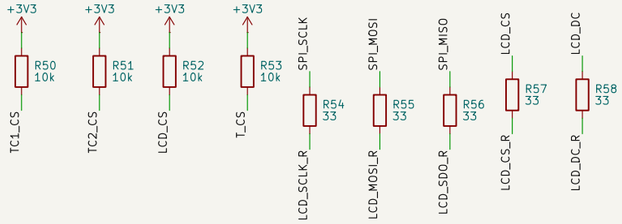


HEADERS
J5 display+touch (14-pin), J6 nav, J7 aux+I2C

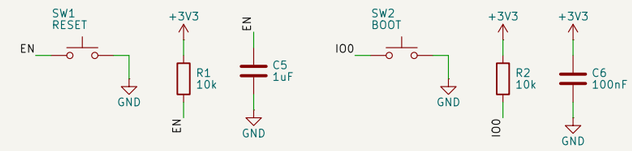


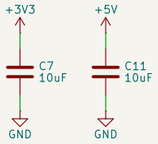
SPI BUS DAMPING + CHIP-SELECT PULL-UPS + I2C EXPANSION
R39-R43 damp the display module's XPT2046 touch lines and
R54-R58 the four 40 MHz display lines; both loom to J5.
R56 is also the SDO hedge - lift it if the panel does not
tri-state J5.9 off the shared SPI_MISO.
R50-R53 hold all four chip selects deasserted from power-on,
before firmware configures any GPIO.
J14 Qwiic and J7 5-8 (0.1 in) share I2C, pulled up by R44/R45



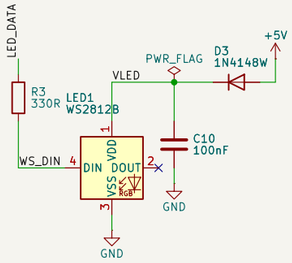


RESET / BOOT / DECOUPLING





WS2812B STATUS LED
VDD dropped ~4.6V for 3.3V data margin



SIGNAL TEST POINTS
SPI, I2C and the two SSR control lines, beside their nets

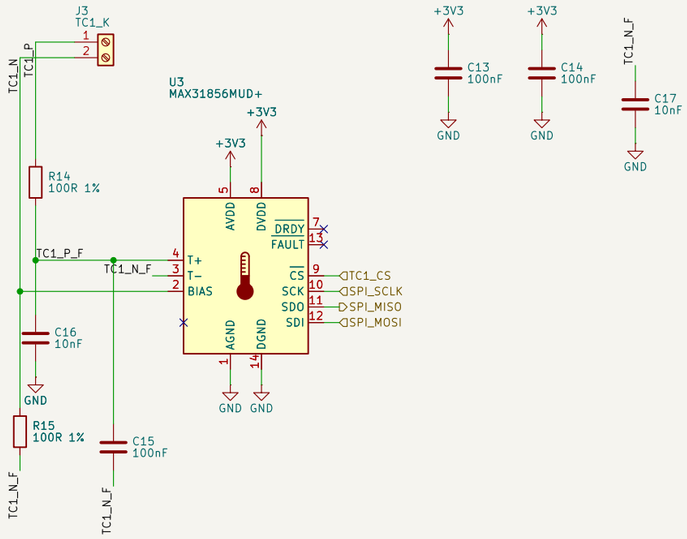


4-layer, 100 x 100 mm, JLCPCB standard process
ESP32-S3-WROOM-1U-N16R2 + 2x MAX31856 + dual SSR + ADE7953
Bisque project
Sheet: /mcu/
File: mcu.kicad_sch

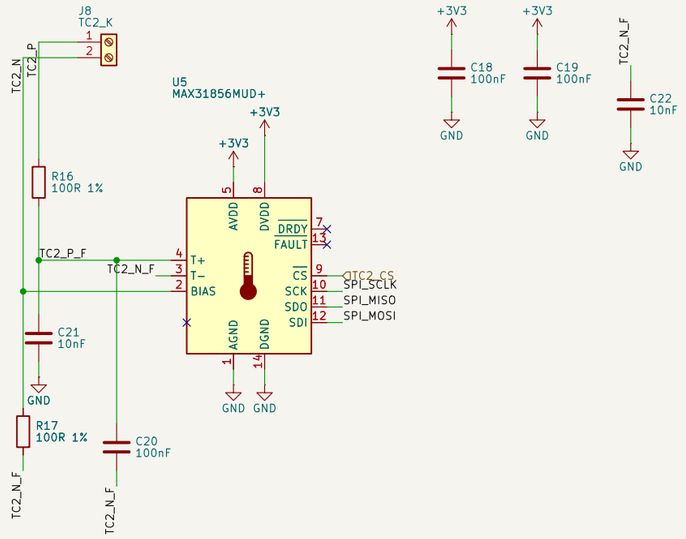
Title: Bisque Kiln Controller — ESP32-S3, USB, headers, status LED

Size: A3	Date: 2026-10-22	**Rev: B.3**
KiCad E.D.A. 10.0.6		Id: 3/8

THERMOCOUPLE 1 (control) MAX31856
T− floats to J3, biased via BIAS



THERMOCOUPLE 2 (load) MAX31856
T− floats to J8, biased via BIAS



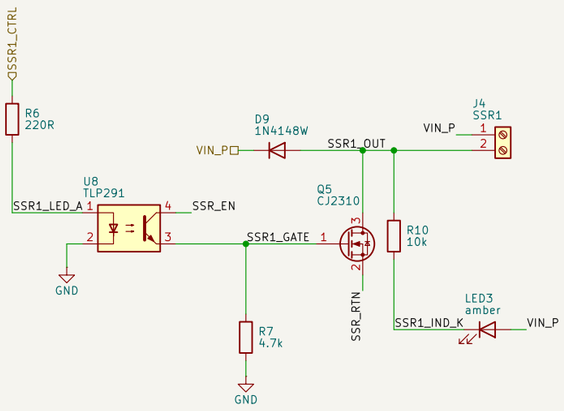
4−layer, 100 x 100 mm, JLCPCB standard process
ESP32−S3−WROOM−1U−N16R2 + 2x MAX31856 + dual SSR + ADE7953
Bisque project

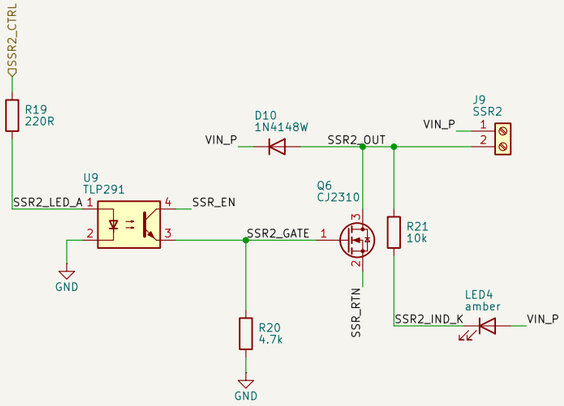
Sheet: /thermocouples/
File: thermocouples.kicad_sch

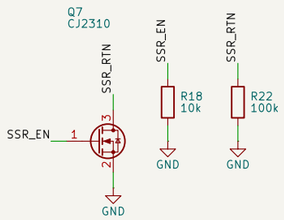
Title: Bisque Kiln Controller − Thermocouple front−ends

Size: A3	Date: 2026−10−22	**Rev: B.3**
KiCad E.D.A. 10.0.6		Id: 4/8

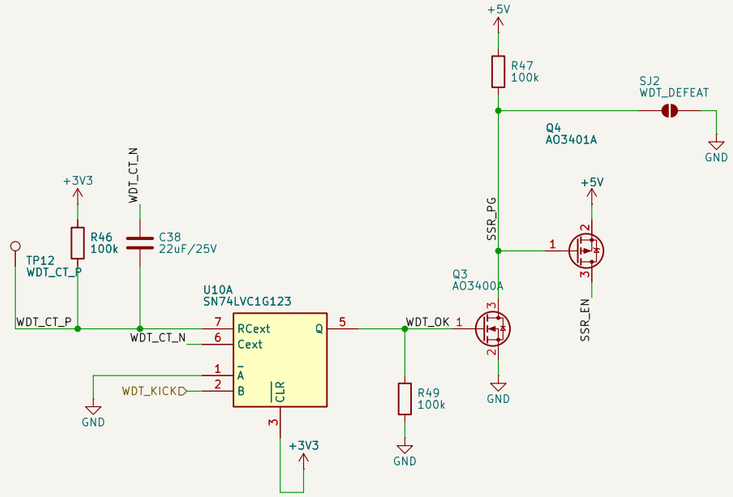
SSR DRIVE x2
24 V terminals off VIN_P, switched low side by a 60 V
CJ2310. The GPIO drives an opto LED, never the gate: at 24 V a
gate–to–drain failure would otherwise put 240 mA into the pin.
Opto collectors sit on SSR_EN (5 V), so the gate lands at ~4.9 V
from the RAIL rather than from CTR. Q7 is the watchdog's gate in
the shared return — it opens a channel whose MOSFET failed short.
R18/R22 define SSR_EN and SSR_RTN when their switches are off.
NOT an isolation barrier — see DESIGN–NOTES.md (SSR block).





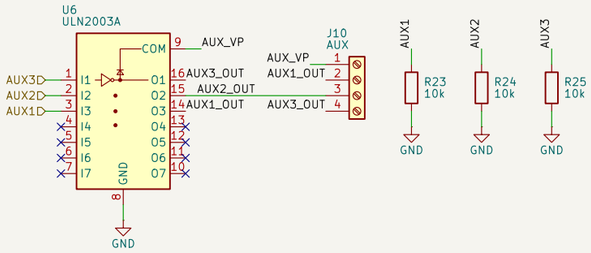


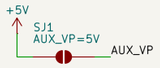
HARDWARE WATCHDOG
U10 retriggerable one–shot: every rising edge on
WDT_KICK restarts a ~2.2 s window, so Q stays high only while
firmware keeps kicking. Q holds Q3 on; Q3 pulls SSR_PG down,
turning on high–side Q4, which supplies SSR_EN — the +5V GATE
DRIVE rail feeding both opto collectors and Q7's gate.
R46/C38 set the window. R48 holds B low when the MCU is high–Z.
R47 is the fail–safe pull–up. SJ2 = bring–up defeat, REMOVE for service.



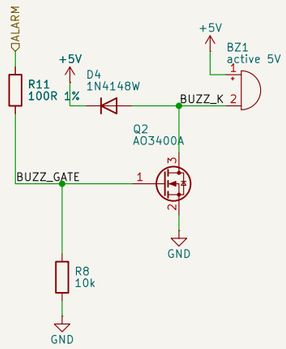
4–layer, 100 x 100 mm, JLCPCB standard process
ESP32–S3–WROOM–1U–N16R2 + 2x MAX31856 + dual SSR + ADE7953
Bisque project
Sheet: /ssr/
File: ssr.kicad_sch
Title: Bisque Kiln Controller — SSR drive and hardware watchdog
Size: A3 | Date: 2026–10–22 | **Rev: B.3**
KiCad E.D.A. 10.0.6 | Id: 5/8

AUX OUTPUT BANK
ULN2003 vent/purge/spare; COM->AUX_VP, SJ1 links to +5V

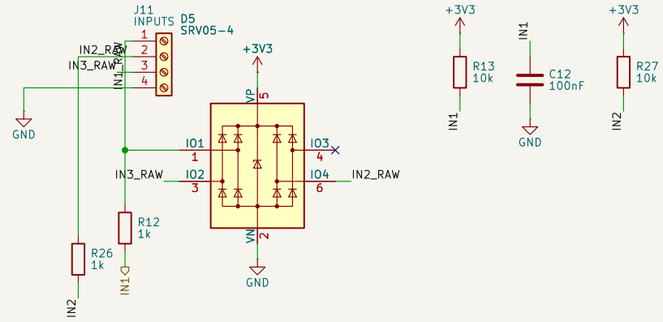


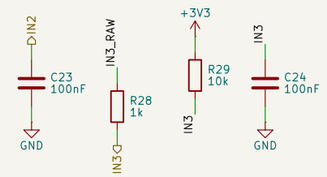


ALARM BUZZER



PROTECTED INPUTS x3 lid / gas-flow / spare
1k + 10k pull-up + 100nF each, SRV05-4 TVS





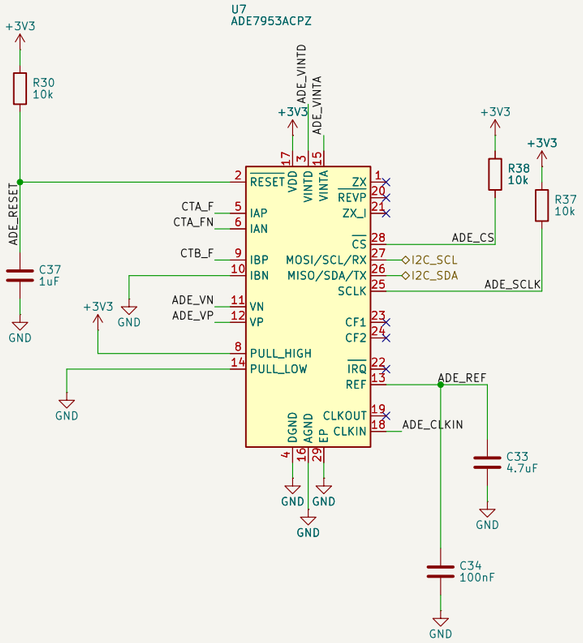
4-layer, 100 x 100 mm, JLCPCB standard process
ESP32-S3-WROOM-1U-N16R2 + 2x MAX31856 + dual SSR + ADE7953
Bisque project

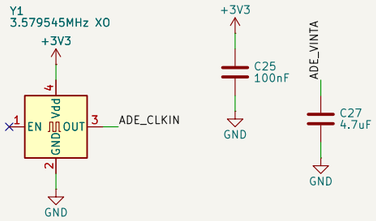
Sheet: /io/
File: io.kicad_sch

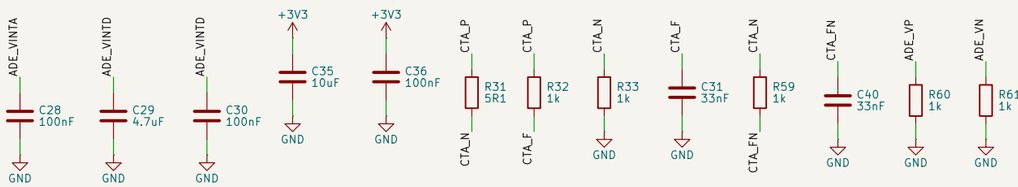
Title: Bisque Kiln Controller — Aux outputs, buzzer, protected inputs

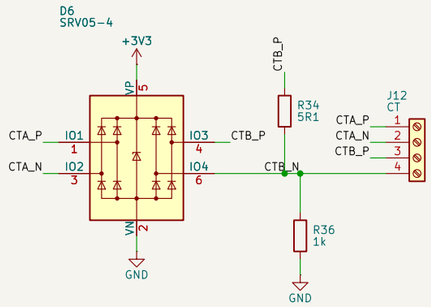
Size: A3	Date: 2026-10-22	**Rev: B.3**
KiCad E.D.A. 10.0.6		Id: 6/8

CT CURRENT SENSING ADE7953, I2C, current-only
no mains – VP/VN to DNP J13













4-layer, 100 x 100 mm, JLCPCB standard process
ESP32-S3-WROOM-1U-N16R2 + 2x MAX31856 + dual SSR + ADE7953

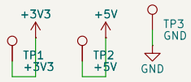
Bisque project

Sheet: /ct/
File: ct.kicad_sch

Title: Bisque Kiln Controller – CT current sensing (ADE7953)

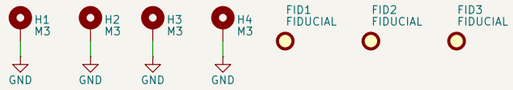
Size: A3	Date: 2026-10-22	**Rev: B.3**
KiCad E.D.A. 10.0.6		Id: 7/8

TEST POINTS



MOUNTING / FIDUCIALS

FID1–3 are bare-copper pick-and-place targets, on no net



4-layer, 100 x 100 mm, JLCPCB standard process
ESP32-S3-WROOM-1U-N16R2 + 2x MAX31856 + dual SSR + ADE7953

Bisque project

Sheet: /test/
File: test.kicad_sch

Title: Bisque Kiln Controller — Test points and mounting

Size: A3	Date: 2026-10-22	**Rev: B.3**
KiCad E.D.A. 10.0.6		Id: 8/8